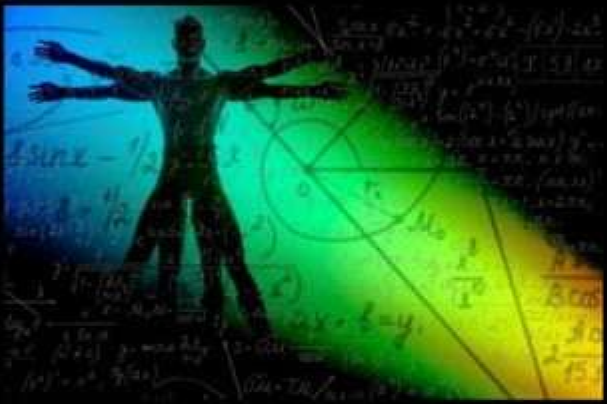
HISTORICAL ANTECEDENTS OF SCIENCE AND TECHNOLOGY

By Prof. Liwayway Memije-Cruz

INTRODUCTION

SCIENCE

- concerted human effort to understand, or to understand better, the history of the natural world and how the natural world works, with observable physical evidence as the basis of that understanding.
- done through observation of natural phenomena, and/or through experimentation that tries to simulate natural processes under controlled conditions.



SCIENCE AND TECHNOLOGY

- Science: knowledge about or study of the natural world based on facts learned through experiments and observation.
- Technology: science or knowledge put into practical use to solve problems or invent useful tools.

HOW IS SCIENCE USED IN TECHNOLOGY?

- Science is the pursuit of knowledge about the natural world through systematic observation and experiments. Through science, we develop new technologies.
- Technology is the application of scientifically gained knowledge for practical purpose.
- Scientists use technology in all their experiments.

THE ROLE OF SCIENCE AND TECHNOLOGY

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1. alter the way people live, connect, communicate and transact, with profound effects on economic development.

2. key drivers to development, because technological and scientific revolutions underpin economic advances, improvements in health systems, education and infrastructure.

3. The technological revolutions of the 21st century are emerging from entirely new sectors, based on micro-processors, tele-communications, bio-technology and nano-technology. Products are transforming business practices across the economy, as well as the lives of all who have access to their effects. The most remarkable breakthroughs will come from the interaction of insights and applications arising when these technologies converge.

4. have the power to better the lives of poor people in developing countries
5. differentiators between countries that are able to tackle poverty effectively by growing and developing their economies, and those that are not.
6. engine of growth
7. interventions for cognitive enhancement, proton cancer therapy and genetic engineering

SOCIETY

- The sum total of our interactions as humans, including the interactions that we engage in to figure things out and to make things
- a group of individuals involved in persistent social interaction, or a large social group sharing the same geographical or social territory, typically subject to the same political authority and dominant cultural expectations.

WHAT DOES SCIENCE TECHNOLOGY AND SOCIETY MEAN?

- Science and technology studies, or science, technology and society studies (STS) is the study of how society, politics, and culture affect scientific research and technological innovation, and how these, in turn, affect society, politics and culture.

SCIENCE AND TECHNOLOGY STUDIES

- STS is a relatively recent discipline, originating in the 60s and 70s, following Kuhn's The Structure of Scientific Revolutions (1962).
- STS was the result of a "sociological turn" in science studies.
- STS makes the assumption that science and technology are essentially intertwined and that they are each profoundly social and profoundly political

HOW SCIENCE AND TECHNOLOGY AFFECT SOCIETY.

- Science and tecnology have had a major impact on society, and their impact is growing.
- By making life easier, science has given man the chance to pursue societal concerns such as ethics, aesthetics, education, and justice; to create cultures; and to improve human conditions.
- Science influences society through its knowledge and world view. Scientific knowledge and the procedures used by scientists influence the way many individuals in society think about themselves, others, and the environment. The effect of science on society is neither entirely beneficial nor entirely detrimental.

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WHAT IS THE RELATIONSHIP BETWEEN SCIENCE AND SOCIETY?

- The impact of science and technology on society is evident. But society also influences science.
- There are social influences on the direction and emphasis of scientific and technological development, through pressure groups on specific issues, and through generally accepted social views, values and priorities

HISTORY OF SCIENCE AND TECHNOLOGY IN THE PHILIPPINES

- Science and technology in the Philippines had experienced periods of intense growth as well as long periods of stagnation.
- The main managing agency responsible for science and technology is the Department of Science and Technology.
- Numerous national scientists have contributed in different fields of science including Fe del Mundo in the field of Pediatrics, Eduardo Quisumbing in the field of Plant taxonomy, Gavino Trono in the field of tropical marine Phycology, Maria Orosa in the field of Food technology and many more

PRE-SPANISH ERA

- Even before the colonization by the Spaniards in the Philippine islands, the natives of the archipelago already had practices linked to science and technology.
- Filipinos were already aware of the medicinal and therapeutic properties of plants and the methods of extracting medicine from herbs.
- They already had an alphabet, number system, a weighing and measuring system and a calendar. Filipinos were already engaged in farming, shipbuilding, mining and weaving.
- The Banaue Rice Terraces are among the sophisticated products of engineering by pre-Spanish era Filipinos.

SPANISH COLONIAL ERA

- The colonization of the Philippines contributed to growth of science and technology in the archipelago.
- The Spanish introduced formal education and founded scientific institution.
- During the early years of Spanish rule in the Philippines. Parish schools were established where religion, reading, writing, arithmetic and music was taught.
- Sanitation and more advanced methods of agriculture was taught to the natives.
- Later the Spanish established colleges and universities in the archipelago including the oldest existing university in Asia, the University of Santo Tomas.

- The Galleon Trade have accounted in the Philippine colonial economy.
- Trade was given more focus by the Spaniard colonial authorities due to the prospects of big profits.
- Agriculture and industrial development on the other hand were relatively neglected.
- The opening of the Suez Canal saw the influx of European visitors to the Spanish colony and some
- Filipinos were able to study in Europe who were probably influenced by the rapid development of scientific ideals brought by the Age of Enlightenment.

AMERICAN PERIOD

- The progress of science and technology in the Philippines continued under American rule of the islands.
- On July 1, 1901 The Philippine Commission established the Bureau of Government Laboratories which was placed under the Department of Interior. The Bureau replaced the Laboratorio Municipal, which was established under the Spanish colonial era. The Bureau dealt with the study of tropical diseases and laboratory projects.
- On October 26, 1905, the Bureau of Government Laboratories was replaced by the Bureau of Science and on December 8, 1933, the National Research Council of the Philippines was established.
- The Bureau of Science became the primary

POST COMMONWEALTH-ERA

- During the 1970s, which was under the time of Ferdinand Marcos' presidency, the importance given to science grew.
- Under the 1973 Philippine Constitution, Article XV, Section 1, the government's role in supporting scientific research and invention was acknowledged.
- In 1974, a science development program was included in the government's Four-Year Development Plan which covers the years 1974-1978.
- Funding for science was also increased.[4] The National Science Development Board was replaced by the National Science and Technology Authority under Executive Order No. 784. A Scientific Career in the civil service was introiduced in 1983.

AMERICAN PERIOD

- Science during the American period was inclined towards agriculture, food processing, forestry, medicine and pharmacy. Not much focus was given on the development of industrial technology due to free trade policy with the United States which nurtured an economy geared towards agriculture and trade.[4]
- In 1946 the Bureau of Science was replaced by the Institute of Science. In a report by the US Economic Survey to the Philippines in 1950, there is a lack of basic information which were necessities to the country's industries, lack of support of experimental work and minimal budget for scientific research and low salaries of scientists employed by the government. In 1958, during the regime of President Carlos P. Garcia, the Philippine Congress passed the Science Act of 1958 which established the National Science Development

POST COMMONWEALTH-ERA

- In 1986, during Corazon Aquino's presidency, the National Science and Technology Authority was replaced by the Department of Science and Technology, giving science and technology a representation in the cabinet.
- Under the Medium Term Philippine Development Plan for the years 1987-1992, science and technology's role in economic recovery and sustained economic growth was highlighted.
- During Corazon Aquino's State of the Nation Address in 1990, she said that science and technology development shall be one of the top three priorities of the government towards an economic recovery.

POST COMMONWEALTH-ERA

- In August 8, 1988, Corazon Aquino created the Presidential Task Force for Science and Technology which came up with the first Science and Technology Master Plan or STMP. The goal of STMP was for the Philippines to achieve newly industrialized country status by the year 2000.
- The Congress did not put much priority in handling bills related to science and technology. The Senate Committee on Science and Technology was one of the committees that handles the least amount of bills for deliberation.

POST COMMONWEALTH-ERA

- Former Science and Technology secretary, Ceferin Follosco, reported that the budget allocation for science and technology was increased to 1.054 billion pesos in 1989 from the previous year's 464 million pesos.
- However, due to the Asian financial crisis, budget allocation for the years 1990 and 1991 were trimmed down to 920 and 854 million pesos respectively. Budget allocation were increased to 1.7 billion pesos in 1992

PARADIGM SHIFTS IN HISTORY

- a typical example or pattern of something.
- a distinct set of concepts or thought patterns, including theories, research methods, postulates, and standards for what constitutes legitimate contributions to a field.

WHAT IS A PARADIGM SHIFT?

- a fundamental change in approach or underlying assumptions.
- a concept identified by the American physicist and philosopher Thomas Kuhn
- a fundamental change in the basic concepts and experimental practices of a scientific discipline. Kuhn presented his notion of a paradigm shift in his influential book The Structure of Scientific Revolutions (1962)

WHY ARE PARADIGM SHIFTS IMPORTANT?

- Paradigm shift is another expression for more significant changes within belief systems.
- Within philosophy of science this concept is sometimes considered important and is sometimes given great attention within education.

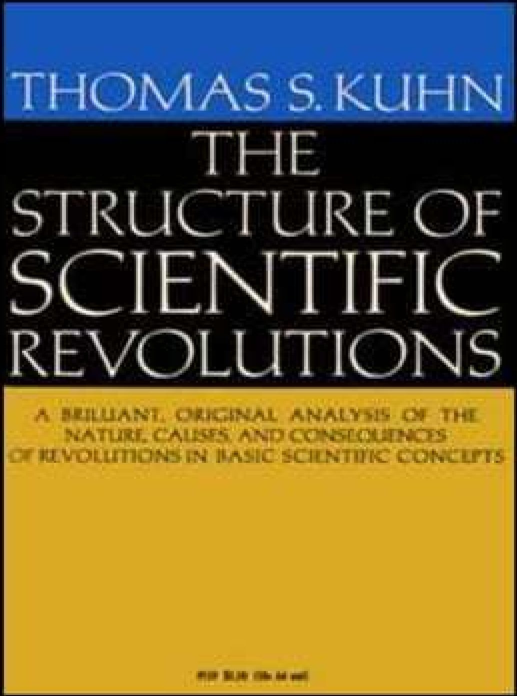
KUHN'S PARADIGM

THOMAS SAMUEL KUHN

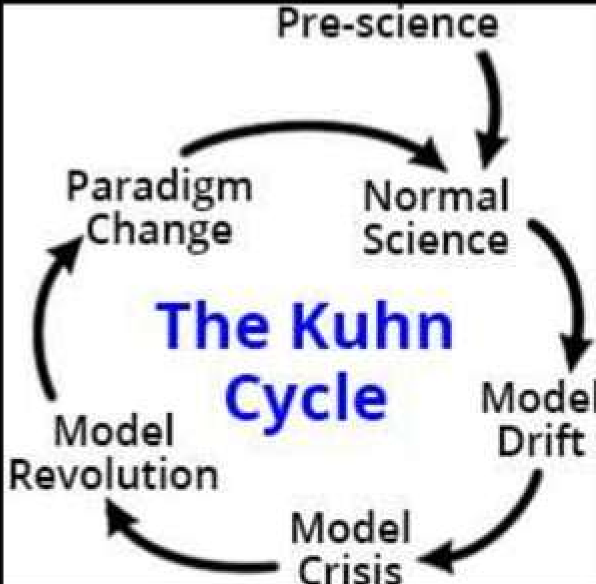
Thomas Samuel Kuhn (/ku:n/; July 18, 1922 – June 17, 1996) was an American physicist, historian and philosopher of science whose controversial 1962 book The Structure of Scientific Revolutions was influential in both academic and popular circles,

THE STRUCTURE OF SCIENTIFIC REVOLUTIONS

- a book about the history of science by the philosopher Thomas S. Kuhn.
- its publication was a landmark event in the history, philosophy, and sociology of scientific knowledge.



THE KUHN CYCLE



- a simple cycle of progress described by Thomas Kuhn in 1962 in his seminal work The Structure of Scientific Revolutions.
- In Structure Kuhn challenged the world's current conception of science, which was that it was a steady progression of the accumulation of new ideas.

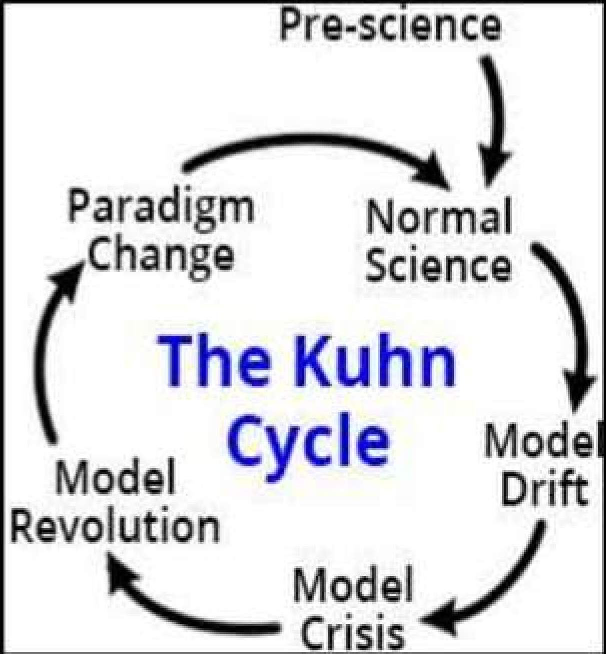
KUHN'S PARADIGM

- Kuhn showed this viewpoint was wrong.
- Science advanced the most by occasional revolutionary explosions of new knowledge, each revolution triggered by introduction of new ways of thought so large they must be called new paradigms.
- Kuhn argues that paradigms change in scientific revolutions. Scientists go through a crisis and transition to a new paradigm, a new way of seeing the world. It is not possible to compare paradigms and it is not possible to say whether one is more right than the other.
- Kuhn argues that science is not moved by a rational process but more by a social unity. In contrast with Popper then Kuhn presents a descriptive theory in which Kuhn try to observe the factual scientific fields in order to understand how they function in practice.

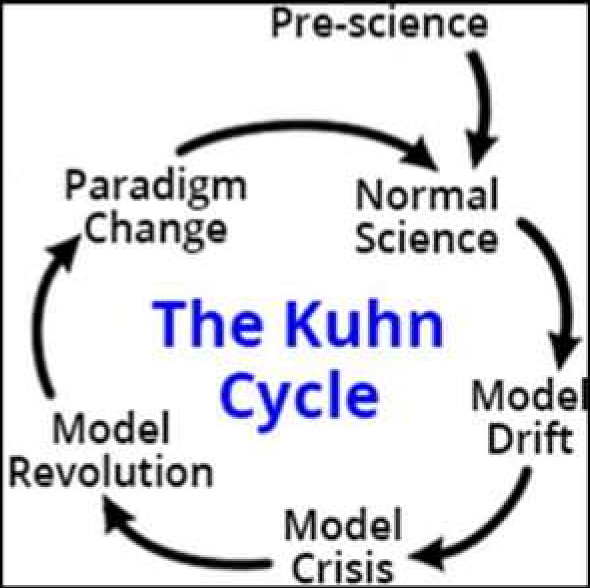
THE STEPS OF THE KUHN CYCLE

PRE-SCIENCE.

- also called the pre-paradigm stage,
- the pre-step to the main Kuhn Cycle. In Prescience there is not yet a model of understanding (the field's paradigm) mature enough to solve the field's main problems.
- The field has no workable paradigm to successfully guide its work.



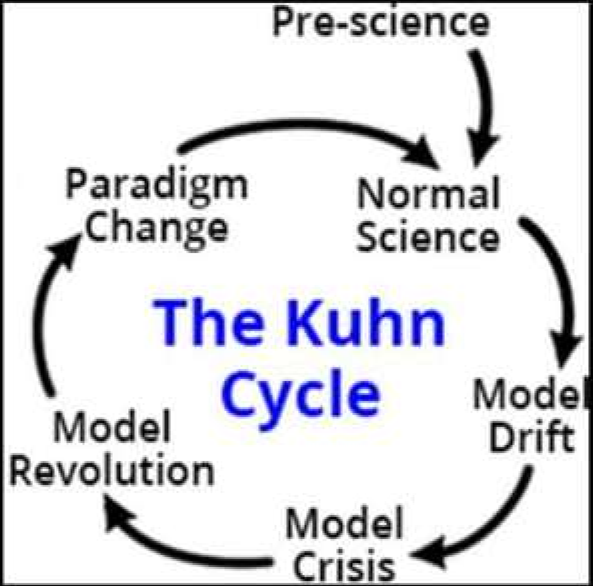
NORMAL SCIENCE



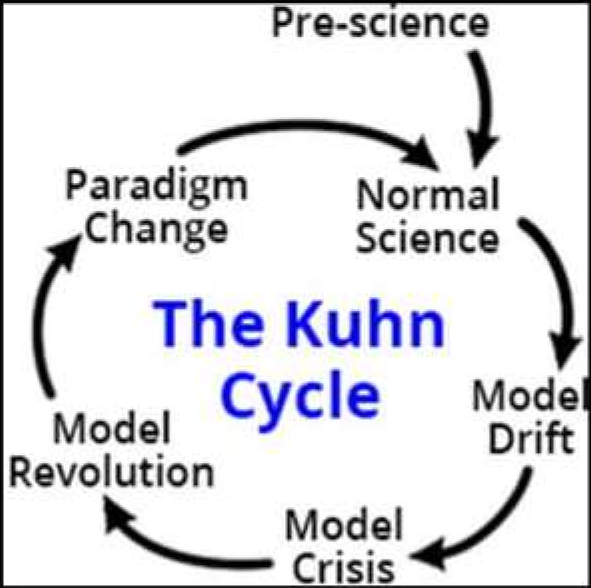
- identified and elaborated on by Thomas Samuel Kuhn in The Structure of Scientific Revolutions, is the regular work of scientists theorizing, observing, and experimenting within a settled paradigm or explanatory framework.
- where the field has a scientifically based model of understanding (a paradigm) that works.

MODEL DRIFT

- The model of understanding starts to drift, due to accumulation of anomalies, and phenomenon, the model cannot explain.



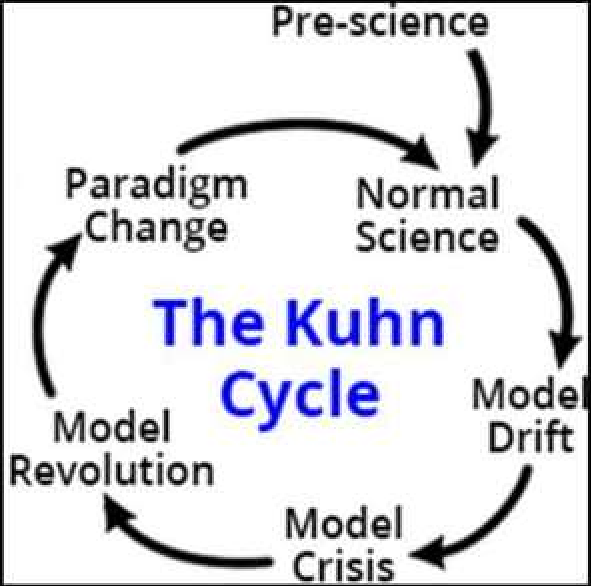
MODEL CRISIS



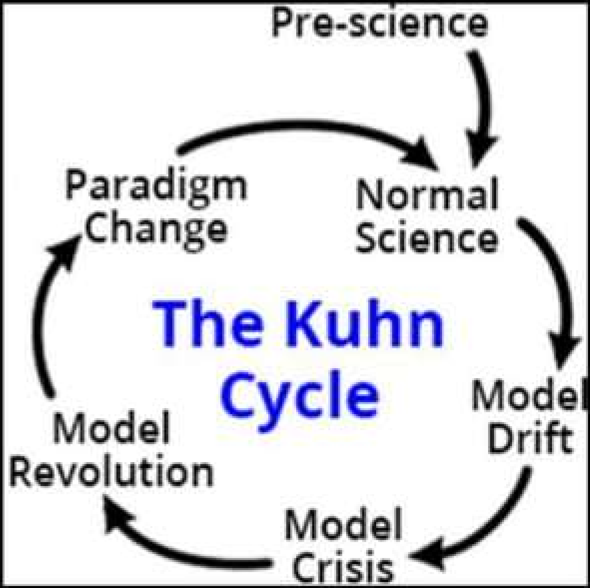
- the most important step of them all in the Kuhn Cycle.
- The Model Drift becomes so excessive the model is broken. It can no longer serve as a reliable guide to problem solving. Attempts to patch the model up to make it work fail. The field is in anguish.

. MODEL REVOLUTION

- begins when serious candidates for a new model emerge. It's a revolution because the new model is so radically different from the old one.
- a field's model of understanding is undergoing revolutionary change. The old model failed, which caused the Model Crisis step. The Model Revolution step begins when one or more competing new models emerge from the crisis.



PARADIGM CHANGE



- also called a paradigm shift,
- Earlier steps have created the new model of understanding (the new paradigm).
- In the Paradigm Change step the new paradigm is taught to newcomers to the field, as well as to those already in it. When the new paradigm becomes the generally accepted guide to one's work, the step is complete. The field is now back to the Normal Science step and a Kuhn Cycle is complete.

VIDEOS/ YOU TUBE

- Stephen Colbert's interview with Neil Tyson
- World's Greatest inventions
- Philippine Great Inventions
- Scientific Reductionism
- What is a Paradigm?

PARADIGM SHIFTS IN HISTORY

HISTORICAL EXAMPLES OF PARADIGMS

SOCIETY / ETHICS

- Slavery is acceptable to now slavery being unacceptable
- Role of Children in Society - Child labor was, now is not acceptable
- ·Male Superiority - Beating wives was, now is not acceptable
- Reading and the Control over information - Invention of the printing press (& other major inventions) allowed for the elites control over reading / writing to end.
- The Reformation- broke monopoly of Catholic Church and Christian's "relationship" with God.

NATURAL SCIENCES

- Darwin's theory of evolution
- Plate Tectonics— create a physical model of the Earth's structure
- Albert Einstein's space-time is not fixed or objective— subject to observer's state of motion relative to other object.

HUMAN SCIENCES

- Psychology: Sigmund Freud—we are not fully in control of our behavior—a subconscious part operate
- Economics—government intervention in economy is now accepted.

THE ARTS

- The Realist paradigm: the purpose of art is to copy reality.
- Shakespeare's impact on drama / theater
- Jazz & rock revolutionizing music

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